Nordic Flow

Allocate goods across existing warehouses at lower total cost

A Sweden-first proposal for Company A · Round 1 · 26 October 2026

8 weeks
from approval to a go / no-go decision

≥ 5%
lower relevant logistics cost: our pilot gate [assumption]

~3–6%
saving needed to break even in year 1 [assumption]

IDENTIFY
Fulfilment is where cost and service meet. We address one decision: which source and existing warehouse serve each order.

PRIORITISE
Chosen over order automation, store standardisation, systems replacement and delivery visibility on impact, speed and data readiness.

SOLVE & SCALE
A planner-approved optimisation service on top of existing systems. Eight-week Swedish pilot; same engine configured for DK and NO.

Goods allocation leads when impact, speed and data readiness are weighed together

Five opportunities scored 1–5 by the team against case evidence; scores are judgement, not company data

Criterion (weight)	A Goods allocation	B Manual order processing	C Store standardisation	D Systems replacement	E Delivery visibility to site
Business impact (30%)	4	3	3	5	5
Fit across DK / NO / SE (25%)	4	3	2	5	3
Time to market (25%)	4	4	2	1	3
Data readiness (20%)	3	3	2	1	2
Weighted score	3.80	3.25	2.30	3.20	3.40

Why A

An internal decision: no new driver, store or customer-facing change.

Uses data the company already holds, so a CSV-based pilot can run in weeks.

Robustness check

If impact is weighted 50%, E ties A at about 3.86 each.

So E is our Phase 2: it reuses the same order and warehouse data.

Why not the others now

B: helps mainly Denmark; most savings claims come from vendors.

C: needs broad discovery in each store.

D: multi-year, high risk.

Weights: impact 30%, fit across three markets 25%, time to market 25%, data readiness 20%. Case pp. 9–12 for the four challenges; delivery visibility added by the team.

The allocation lever is plausible and modest, so Week 1 measures it before we commit

Separating what the case confirms from what we must verify

Confirmed in the case	Hypotheses to test in Week 1	Outside evidence (other settings)
• Hundreds of stores in Denmark, Norway and Sweden, each market with its own systems. • Fulfilment and fragmented systems affect efficiency and reliability. • Impact and urgency differ by market.	• Planners optimise one leg at a time, not total cost. • Rates, stock and capacity sit in separate local systems. • Stock records may be wrong, so plans send goods to warehouses that do not have them.	• 4.8% lower delivered cost, building-products maker; modelled [cited] • 10–15% logistics saving, 150+ warehouse network; realised [cited, consultancy] • 65% of stock records inaccurate; profit down >10%, one US retailer [cited] • Byggmax restructured logistics "to increase control over flows" [cited]

Reading: gains of a few percent are plausible, not guaranteed. We therefore gate the pilot on a measured cost gap, not on a promised saving.

Sources: Case brief pp. 8–11; David et al., Interfaces 45(3), 2015; McKinsey distribution-network case; DeHoratius & Raman, Management Science 54(4), 2008; Byggmax Annual Report 2025, p. 6.

A focused pilot: one Swedish region, three warehouses, 20 SKUs, one weekly planning run

Scope is the smallest network that still contains a real routing choice; all sizes are assumptions Company A must confirm

2	3	20	1
suppliers	existing warehouses	palletised SKUs	weekly planning run

In scope

- Allocate inbound supply and confirmed orders across existing warehouses
- Compare the current allocation with a feasible lower-cost plan, with cost by leg
- Planner approves; every override is logged

Out of scope

- New warehouses or site selection
- Vehicle routing (stays in the current TMS)
- Demand forecasting, production planning
- ERP replacement; other countries

Why Sweden first: the case ties Sweden to fulfilment impact. It is a discovery candidate, not a commitment: if Week 1 data is better in Denmark or Norway, the pilot market changes.

Sizes are [assumption]. Byggföretagen forecast: building investment +6% in total 2026–27, a fragile recovery [cited, autumn 2026 update]; case p. 10 lists delivery and fulfilment as Sweden's priority.

The planner stays in control: the service recommends, a person approves, existing systems execute

The recommendation fits today's process instead of replacing it

01 Load

Orders, available supply, rates and warehouse limits from existing systems

›

02 Compare

Current allocation versus a feasible lower-cost plan

›

03 Review

Exceptions, service rules and cost by leg

›

04 Approve

Planner exports the plan; ERP / WMS / TMS execute it

Recommendation output (example)

Supplier / SKU	Warehouse	Destination	Quantity / required date
S1 / SKU-01	Warehouse A	Order zone North	60 pallets / within the week
S2 / SKU-01	Warehouse B	Order zone South	40 pallets / within the week

Not in the pilot

- ERP replacement
- Automatic execution of plans
- Replacing planner judgement

Design principle: reuse before build. Local systems stay authoritative; we add one thin layer.

Illustrative output only. Each run stores the input version, solver status, approvals and planner overrides.

A constrained optimisation engine does the maths; AI helps around it, always with a human approving

Optimisation research, not a black box: every constraint and solver status is visible

Decision engine

Minimise inbound freight + handling + outbound freight + storage fees that change with the plan

Subject to demand met, supply limits, warehouse handling limits, SKU eligibility, allowed lanes

LP for divisible quantities; **MILP** for whole pallets and discrete commercial rules

Solver status shown: optimal, feasible within time limit, or infeasible with a diagnosis (short on supply, capacity or lanes)

Fixed rents already committed are not counted as savings.

Where AI adds value

1 Data mapping. Suggests SKU and unit matches across country systems; a person validates.

2 Plain-language explanations. Why a warehouse changed and which constraint binds, so the planner can check quickly.

3 Learning from overrides (Phase 2). Planner corrections refine rules over time.

Responsible by design: goods allocation is not an Annex III high-risk use [cited]; no profiling of staff; data stays in EU hosting.

EU AI Act, Regulation (EU) 2024/1689, Art. 6 and Annex III. Google OR-Tools is open source (Apache 2.0).

A worked example shows where the saving comes from: total cost across both legs

Synthetic data: 1 SKU, 2 source warehouses, 2 destination zones; cost units per pallet; both plans serve the same 100 pallets

Source	A / North	A / South	B / North	B / South
S1	5	9	11	8
S2	6	10	8	5

Supply S1 / S2 = 60 / 40. Demand North / South = 60 / 40. Warehouse capacity A / B = 100 / 60.

Plan	Allocation	Total cost
Baseline: preferred warehouse A	60 via S1–A–North + 40 via S2–A–South	700
Optimised	60 via S1–A–North + 40 via S2–B–South	500

700

baseline

500

lowest feasible cost

What it shows

- The saving (200 units, 28.6%) comes from seeing both legs together, not from a cheaper single leg.
- Habit fills the preferred warehouse; the optimiser uses the cheaper route that still respects capacity.
- Real gains are expected to be far smaller than 28.6%; see the next slide.

Synthetic example for illustration; it demonstrates the mechanism and is not a forecast for Company A.

The case to proceed: a saving of only ~3–6% breaks even in year one, and Week 1 measures the real figure

We do not promise ROI; we show what must be true and gate the pilot on it

Annual saving = relevant logistics cost in pilot scope × saving rate

Break-even rate = (build ≈ €0.2M + run ≈ €0.1M) ÷ relevant cost

Relevant cost in scope	Saving at 2%	at 5%	at 10%	Break-even
€5M / year	€0.10M	€0.25M	€0.50M	6.0%
€10M / year	€0.20M	€0.50M	€1.00M	3.0%

Reading the table

- Only costs that change with allocation count; committed warehouse rents do not.
- The 5% gate clears break-even when the relevant cost is about €6M or more; Week 1 confirms the base.
- If Week 1 finds a smaller cost base or gap, we stop.

Pilot acceptance gates

Measure	Gate (hypothesis)
Relevant logistics cost	≥ 5% below a validated feasible baseline, same orders
Planner preparation time	≥ 30% lower, full task incl. data checks
Service	No deterioration in on-time, in-full rate
Feasibility	Zero hard-constraint violations

Scale only when validated savings cover delivery and operating cost. No revenue uplift is assumed.

All cost inputs are [assumption], replaced with Company A data in Week 1. Outside benchmarks: 4.8% modelled (Interfaces, 2015), 10–15% realised (McKinsey). Labour savings assessed separately.

One engine, three rollouts: each market gets the same core with local data and rules

The case: one strategic solution does not mean one identical implementation

Wave	Market	What changes locally	Evidence to collect first
1	Sweden	Pilot one region from CSV exports; planner reviews every plan	Rates, stock, service rules, warehouse capacity
2	Denmark	Reuse the engine; adapt the import to cut manual preparation	Structured B2B e-invoices only 5–10% [cited], so inputs need more cleaning; wholesale EDI sales 39% [cited]
3	Norway	Validate branch and distribution workflows before connecting	Structured B2B e-invoices 89% [cited]; wholesale EDI sales 27% [cited]; store receiving limits, carrier rules

Shared core

Data contract, optimisation rules, audit trail, KPI definitions.

Local teams own rates, service rules and approval rights.

Employees

Planners co-design the workflow and keep the final say; overrides are logged, not punished. Training in the first week of each wave.

Customers

No change to ordering. On-time, in-full rate is a gate before each wave, so service cannot quietly worsen.

Eurostat isoc_ec_eseln2 (2024, wholesale, ≥10 employees); Digdir (NO) and Erhvervsstyrelsen (DK) e-invoicing analyses. Rollout order is a proposal, not a finding about local systems.

Eight weeks to a pilot decision, with the main risks designed out and a clear ask

Indicative plan after sponsor approval and access to usable data

Week 1

Validate problem and data

Exit: Agreed baseline, scope, service rules

Weeks 2–4

Build MVP: engine, import, plan view

Exit: Plans pass historical replay

Weeks 5–6

Controlled trial with local planners

Exit: Planner acceptance, service evidence

Weeks 7–8

Review economics and readiness

Exit: Scale, revise or stop against gates

Risk	Mitigation
Stock or rate data is wrong (65% of records in one study)	Validation step; block invalid runs; planner approves
Inbound freight is paid by suppliers (seen at Byggmax)	Week 1 confirms who pays; if suppliers, optimise warehouse-to-customer leg and handling only
Weak baseline overstates the gain	Compare against a validated feasible baseline on the same orders
Low planner adoption	Co-design; keep overrides; show cost by leg

Decision requested

Approve Week 1 Swedish discovery and a capped pilot design.

Name a business owner and a local process owner.

Proceed only if data, a measured cost gap and ownership are confirmed; otherwise iterate or stop.

Stock-record accuracy: DeHoratius & Raman (2008). Plan and sizes are indicative and depend on data access.